PHOT 451: Microscale optical system design
LECTURE 6

Michaël Barbier, Fall semester (2026-2027)

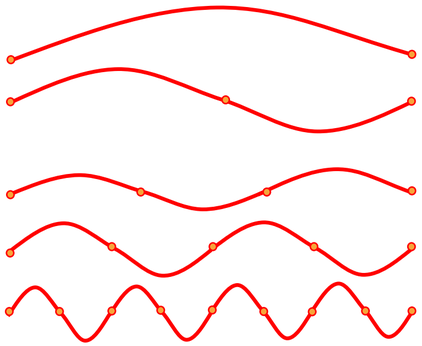
OVERVIEW OF THE COURSE

week	Topic
Week 1	Introduction to micro-scale optical components
Week 2	Light propagation in free space
Week 3	Geometric optics and raycasting
Week 4	Diffraction limit & Abberations
Week 5	Beam propagation
Week 6	Refractive optical elements Microlenses
Week 7	**Midterm** + Blazed Fresnel lenses
Week 8	Digital lenses
Week 9	Diffractive optical elements
Week 10	Wave guides and beam propagation
Week 11	Wave mixing
Week 12	**Project presentations**
Week 13	photonic crystals, Gratings, periodic structures
Week 14	Whole optical system optimization

Light as information

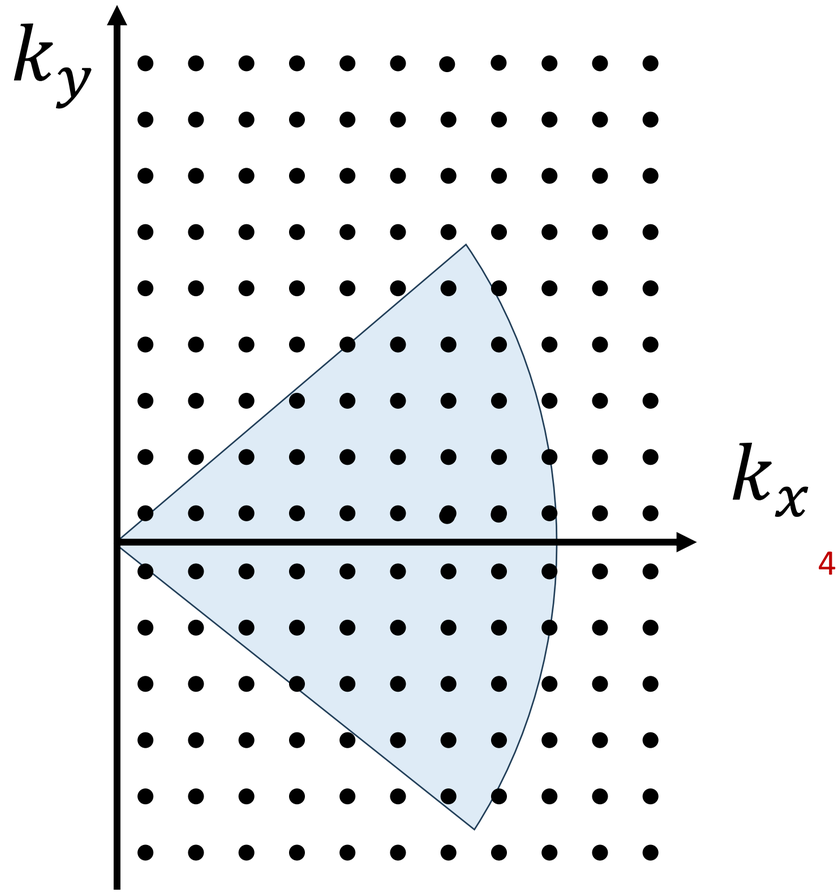
LIGHT AS INFORMATION CARRIER

- Frequency range:
  - Frequency range carries more information
  - Resolution of imaged objects



$$\sigma_{k_x}^2 = \frac{\int_{-\infty}^{+\infty} (k_x - k_0)^2 \, |\tilde{u}(k_x)|^2 \, dk_x}{\int_{-\infty}^{+\infty} |\tilde{u}(k_x)|^2 \, dk_x}$$

$$k_0 = \frac{\int_{-\infty}^{+\infty} k_x |\tilde{u}(k_x)|^2 \, dk_x}{\int_{-\infty}^{+\infty} |\tilde{u}(k_x)|^2 \, dk_x}$$



LIGHT AS INFORMATION CARRIER

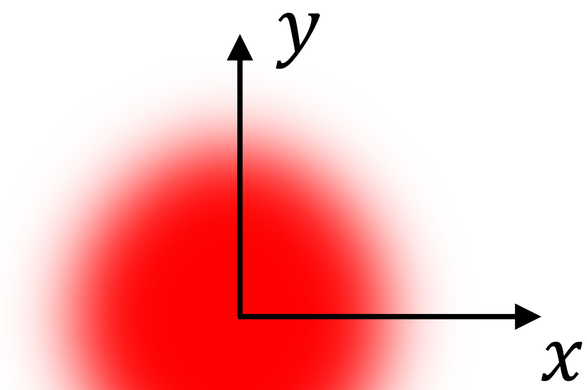
- Complex amplitude optical field $u(x)$

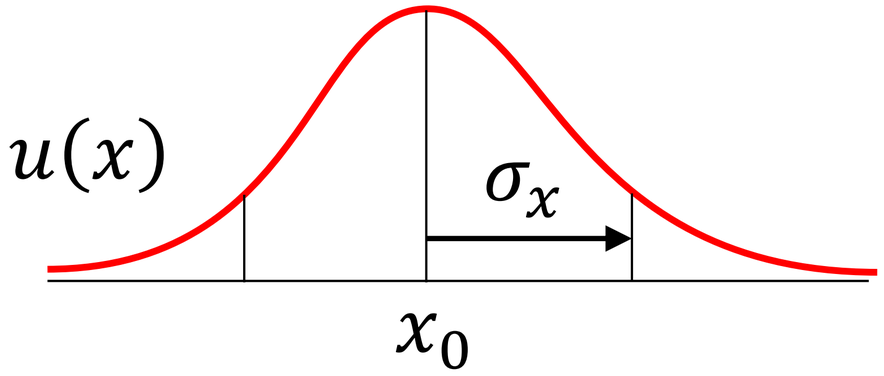
- Size of the object:

$$\sigma_x^2 = \frac{\int_{-\infty}^{+\infty} (x - x_0)^2 \, |u(x)|^2 \, dx}{\int_{-\infty}^{+\infty} |u(x)|^2 \, dx}$$

$$x_0 = \frac{\int_{-\infty}^{+\infty} x \, |u(x)|^2 \, dx}{\int_{-\infty}^{+\infty} |u(x)|^2 \, dx}$$

Example: Gaussian beam





LIGHT AS INFORMATION CARRIER

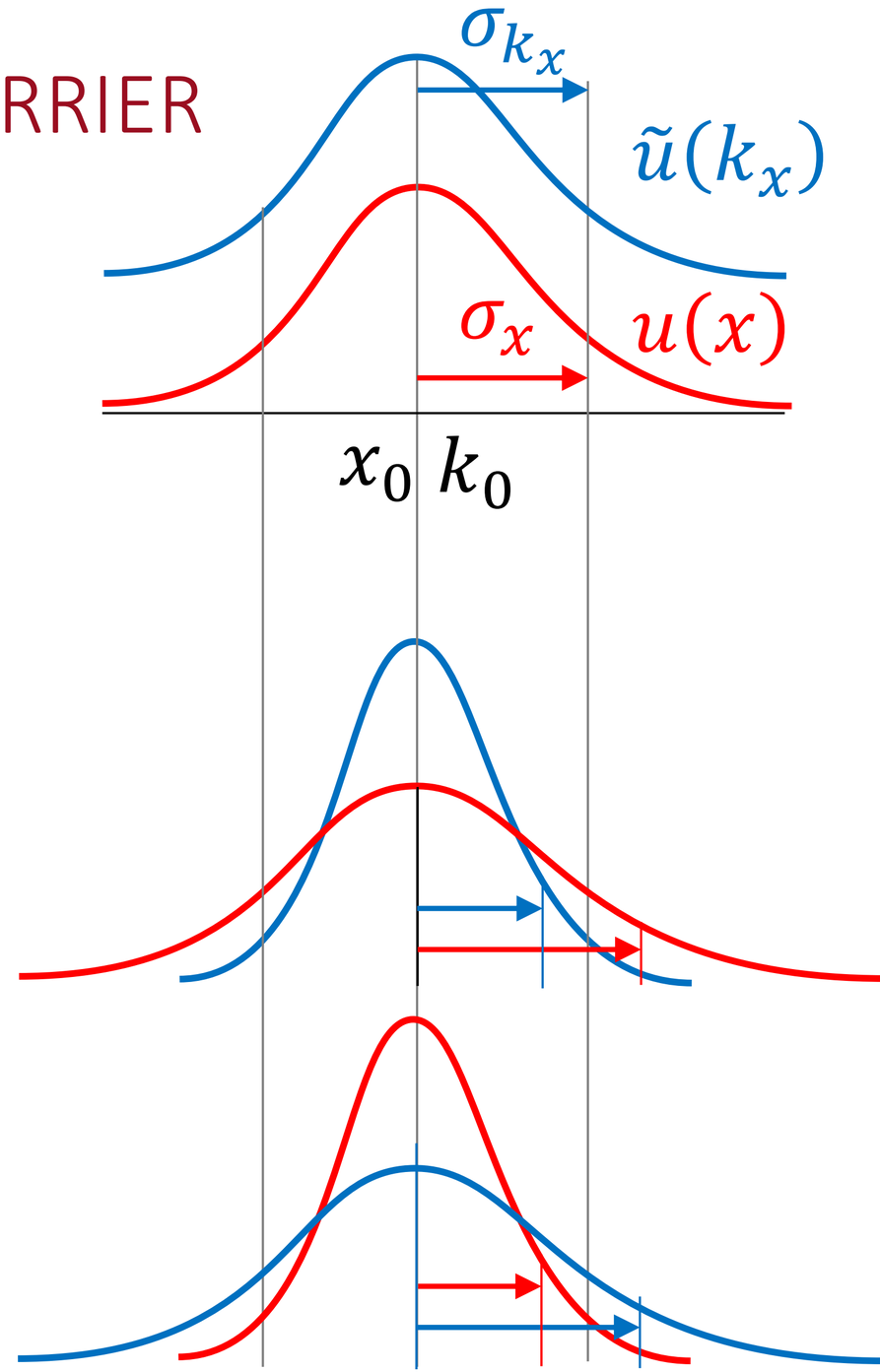
- Space Bandwidth Product (SBP)

$$\mathrm{SBP}_{1\mathrm{D}} = \sigma_x \, \sigma_{k_x}$$

- Minimum SBP for Gaussian $u(x) = e^{-\frac{x^2}{w^2}}$:

$$\mathrm{SBP}_{1\mathrm{D}} = \sigma_x \, \sigma_{k_x} \geq \frac{1}{2}$$

- Similar to Heisenberg uncertainty principle in quantum mechanics



LIGHT AS INFORMATION CARRIER

- Quality factor and SBP

$$M^2 = \mathrm{SBP}_{1\mathrm{D}} = \sigma_x \ \sigma_{k_x} \geq \frac{1}{2}$$

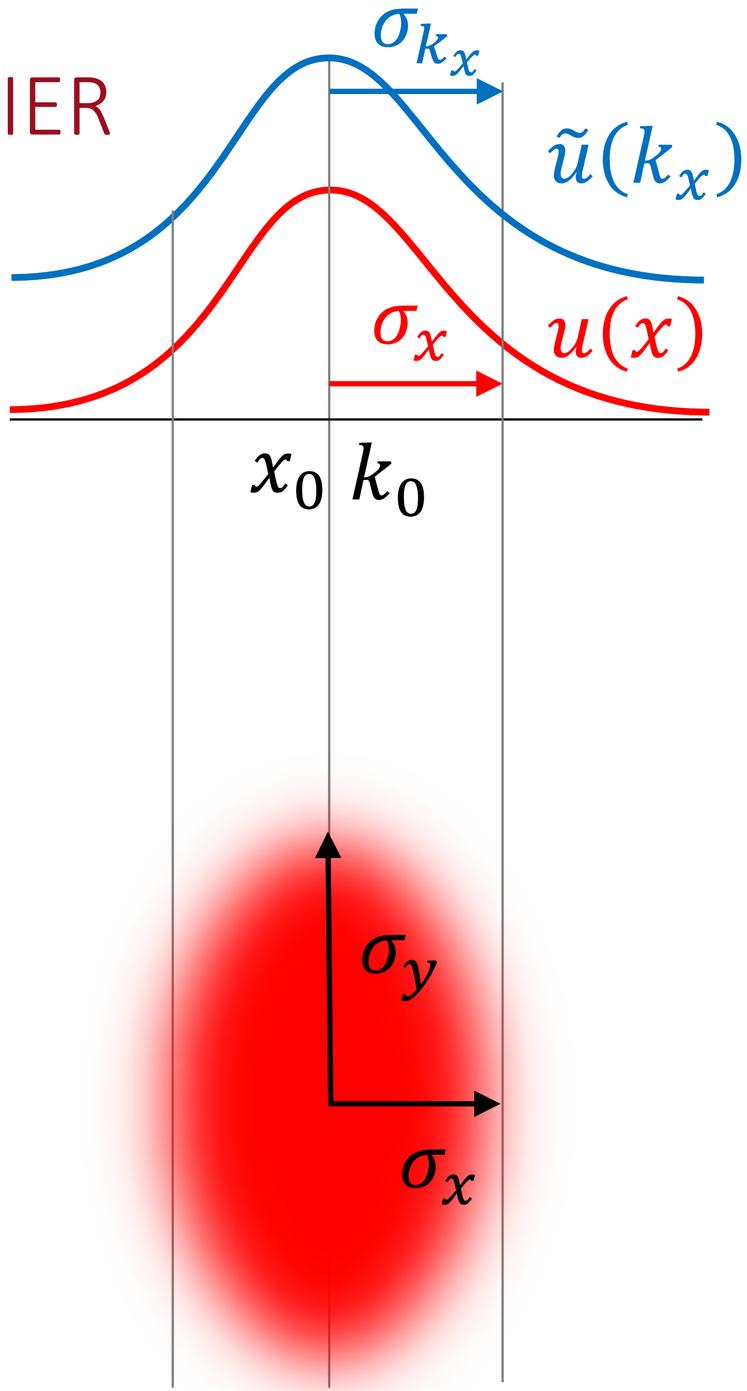
- In two dimensions:

$$\mathrm{SBP}_{2\mathrm{D}} = M^4 = \sigma_r^2 \sigma_k^2$$

- Quality **astigmatic** Gaussian beam

$$\mathrm{SBP}_{2\mathrm{D}} = M_x^2 M_y^2 = \sigma_x \sigma_{k_x} \cdot \sigma_y \sigma_{k_y}$$

$$= \frac{\left(w_x^2 + w_y^2\right)^2}{2 w_x w_y} \geq 1$$



Diffraction limit and resolution

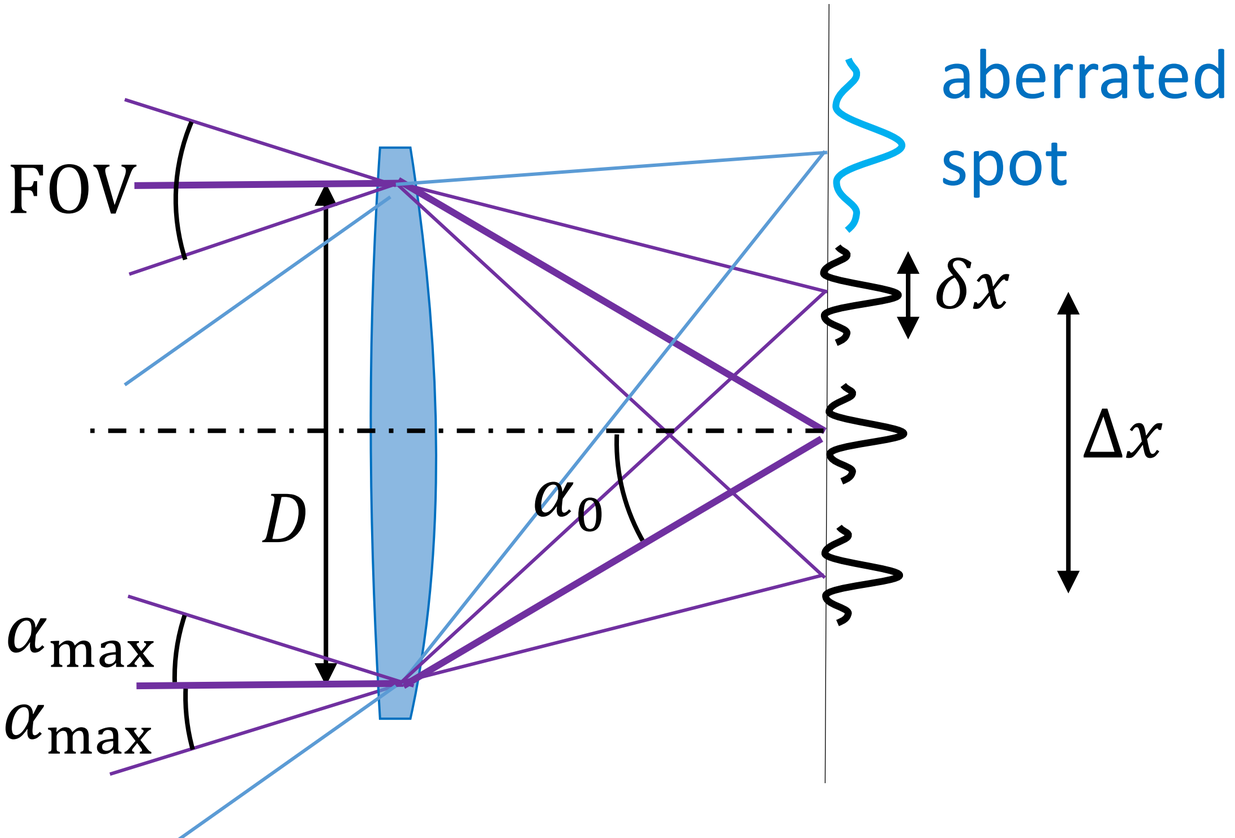
NUMERICAL APERTURE

- Numerical aperture NA

$$\mathrm{NA} = \frac{D}{2f}$$

- Small aberrations:

$$\mathrm{FOV} = 2\alpha_{\max}$$



NUMERICAL APERTURE

- Numerical aperture NA

$$\mathrm{NA} = \frac{D}{2f}$$

$$\Delta x = 2f\,\tan(\alpha_{\max}) \approx f \cdot \mathrm{FOV}$$

$$\delta x \approx 2.44\frac{\lambda f}{D} = 1.22\frac{\lambda}{\mathrm{NA}} = 1.22\lambda\,\tan\alpha_0$$

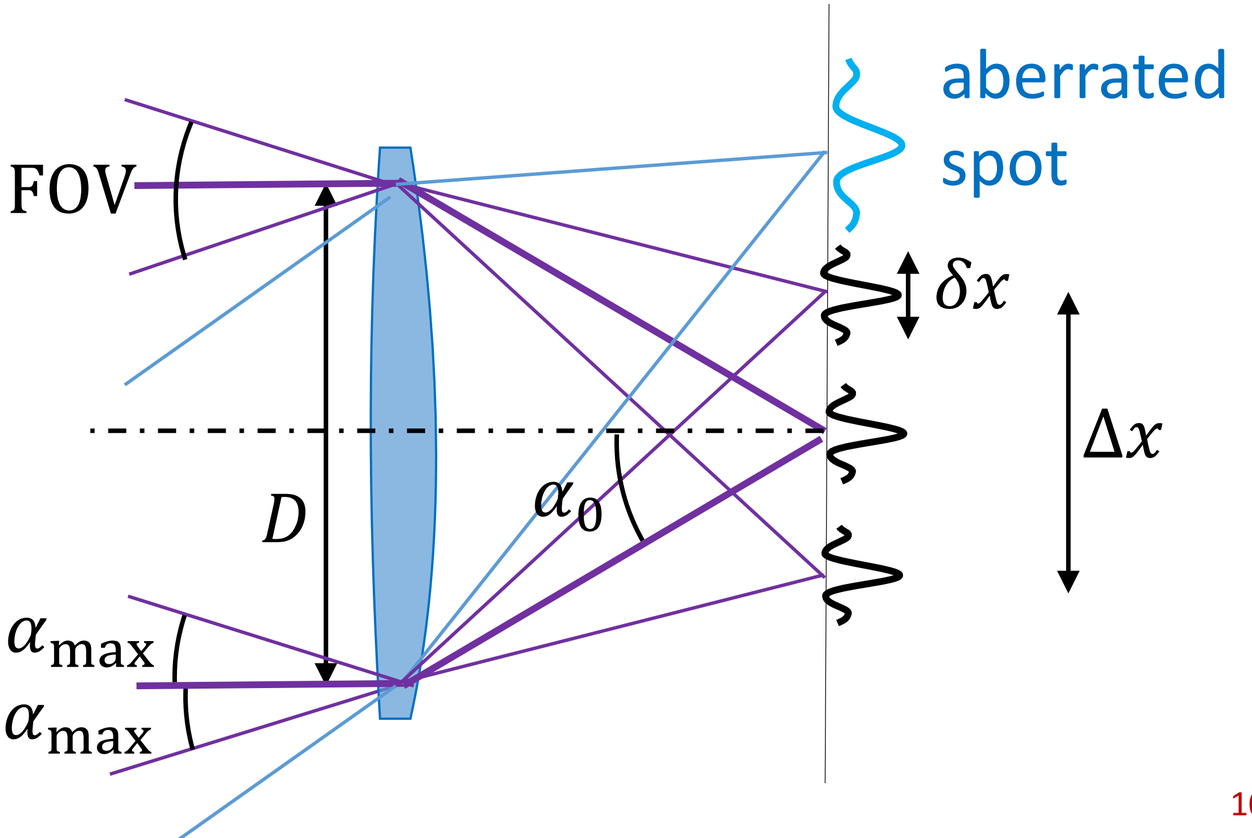
- Small aberrations:

$$\mathrm{FOV} = 2\alpha_{\max}$$

- Space bandwidth product

$$\mathrm{SBP_{1D}} = \frac{\Delta x}{\delta x}$$

$$\approx 1.22\frac{f}{\lambda}\ \mathrm{FOV} \cdot \mathrm{NA}$$



Downscaling of an optical system

DOWNSCALING OPTICAL SYSTEMS

- Assume scaling parameter $\mathcal{M}$
- Option 1: **constant f/# scaling**
  - Focal length $f \rightarrow \mathcal{M}f$
    Lens diameter $D \rightarrow \mathcal{M}D$

Scale both lens diameter D and surface radii R_1 and R_2

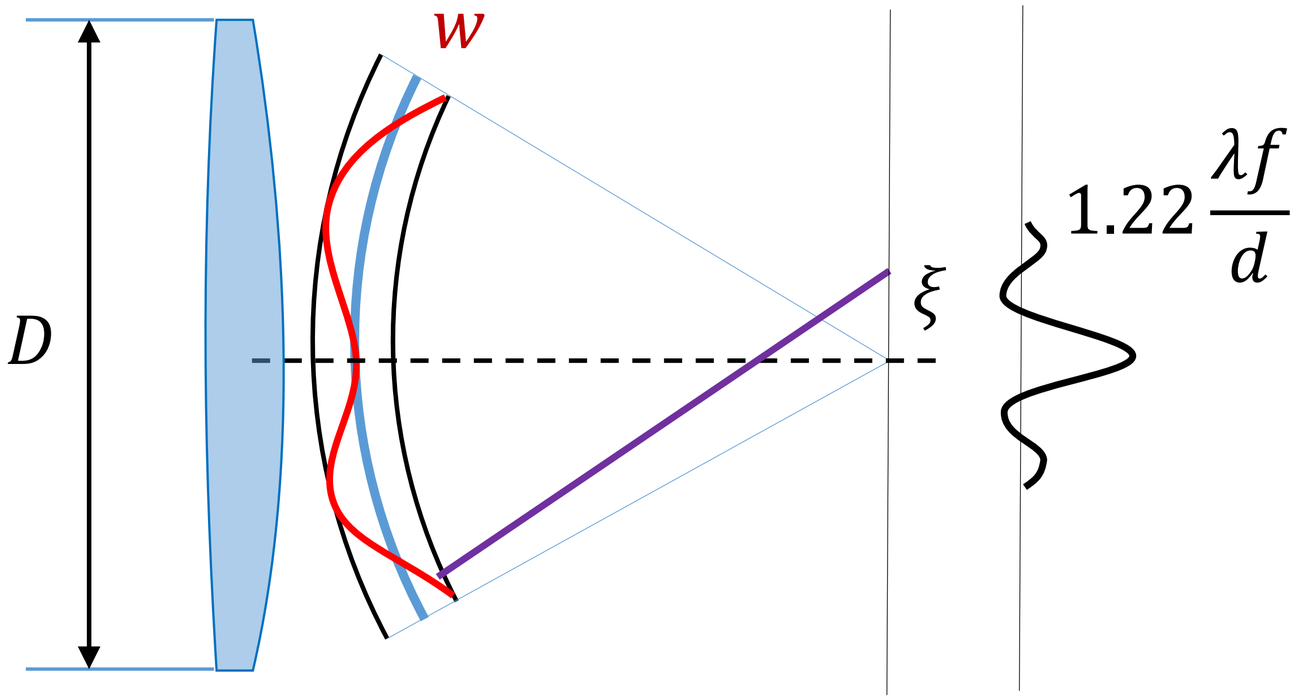
$$\frac{1}{f} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

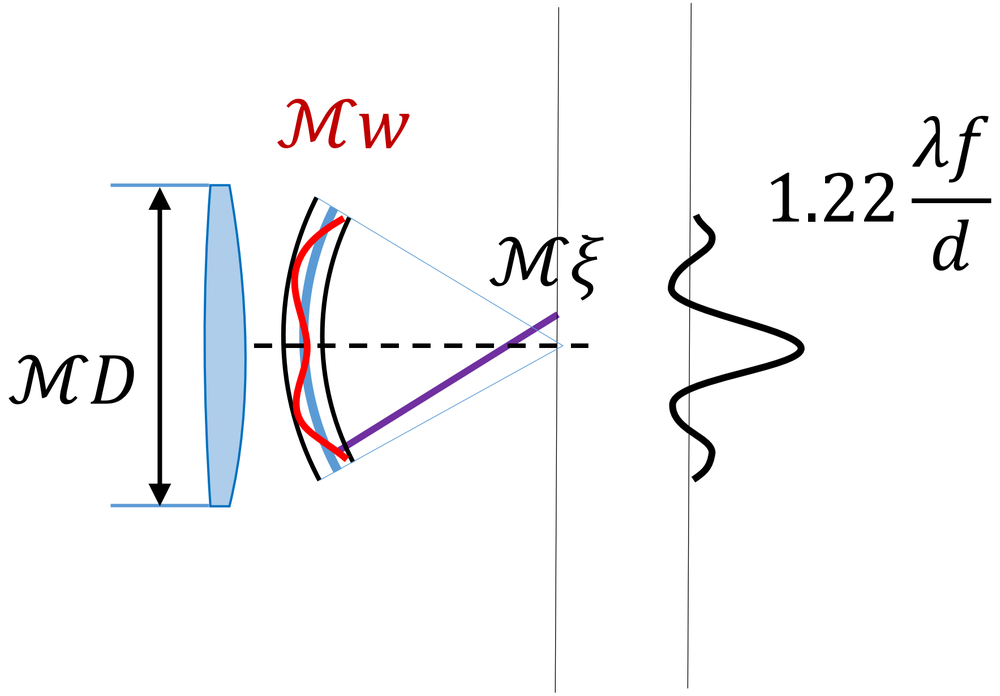
DOWNSCALING OPTICAL SYSTEMS

- Assume scaling parameter $\mathcal{M}$
- Option 1: **constant f/# scaling**
  - Focal length $f \rightarrow \mathcal{M}f$
    Lens diameter $D \rightarrow \mathcal{M}D$

Scale both lens diameter D and surface radii R_1 and R_2

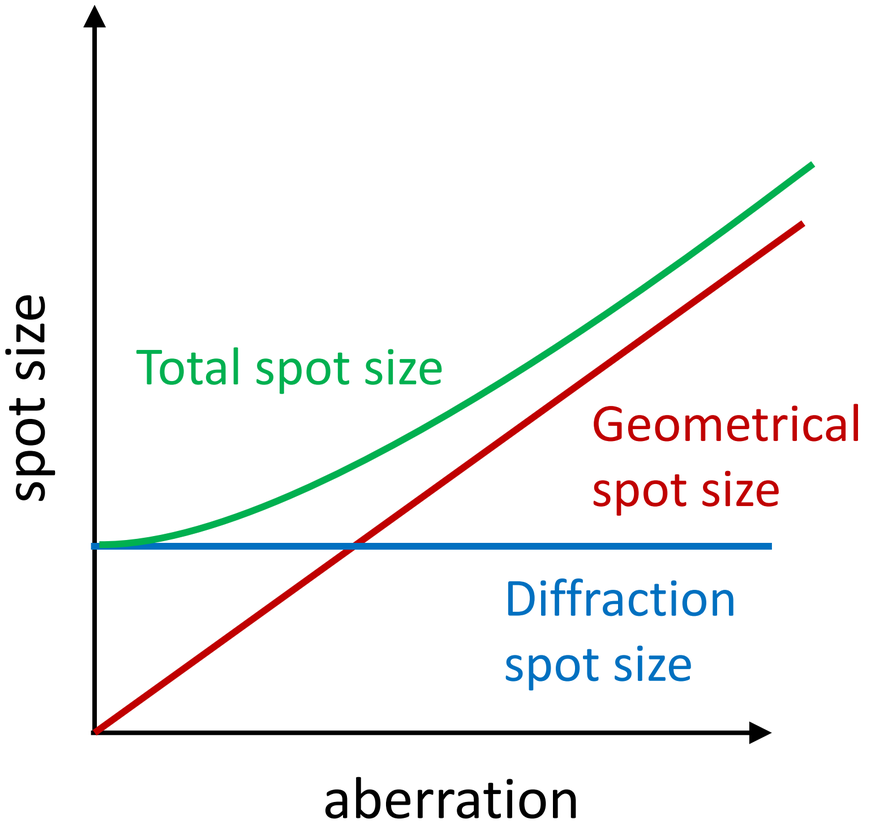
$$\frac{1}{f} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$



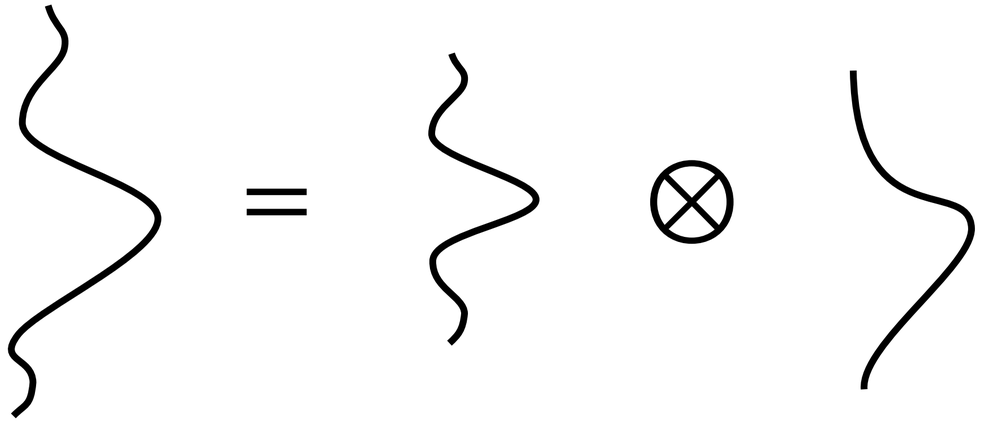


DOWNSCALING OPTICAL SYSTEMS

- **Constant f/# scaling**
  - Focal length $f \rightarrow \mathcal{M}f$
  - Lens diameter $D \rightarrow \mathcal{M}D$



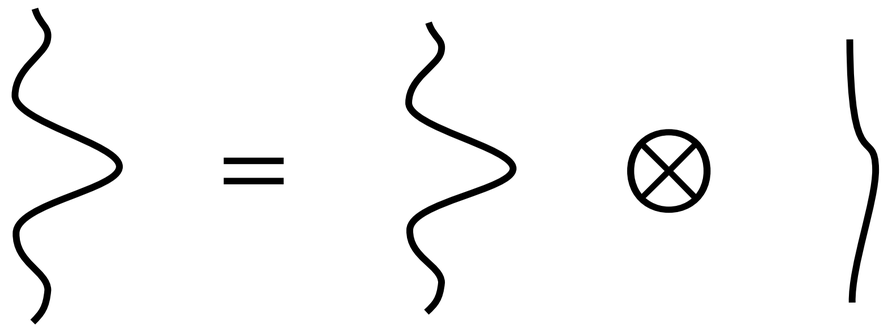
$$\sigma_{\text{tot}}^2 = \sigma_{\text{psf}}^2 + \sigma_{\text{aberration}}^2$$

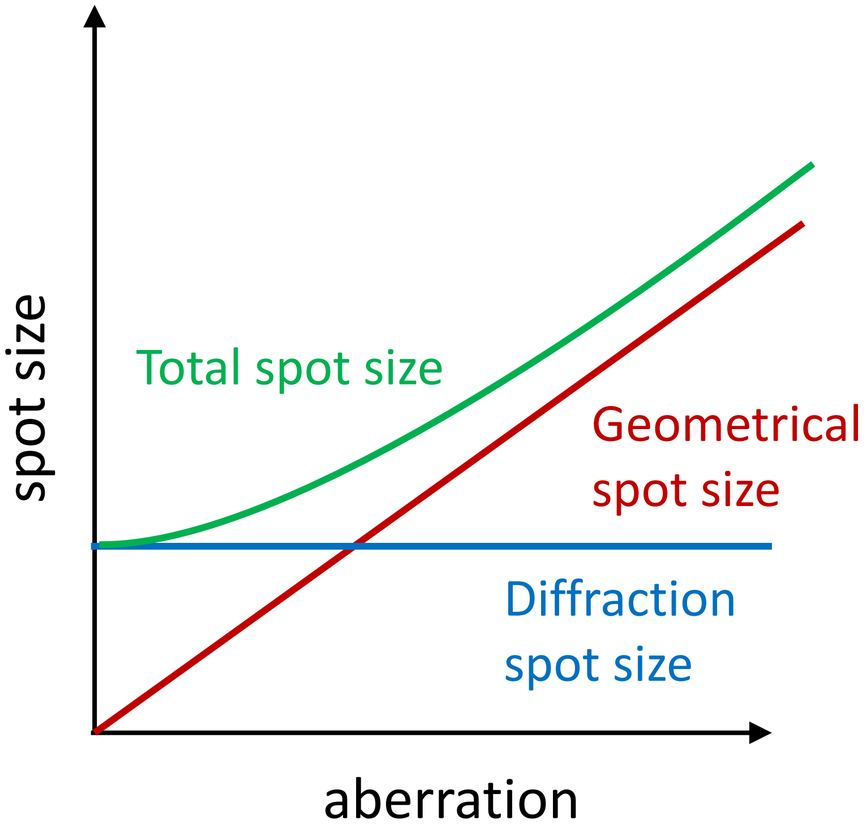


DOWNSCALING OPTICAL SYSTEMS

- **Constant f/# scaling**
  - Focal length $f \rightarrow \mathcal{M}f$
  - Lens diameter $D \rightarrow \mathcal{M}D$

$$\sigma_{\text{tot}}^2 = \sigma_{\text{psf}}^2 + \mathcal{M}^2\sigma_{\text{aberration}}^2$$

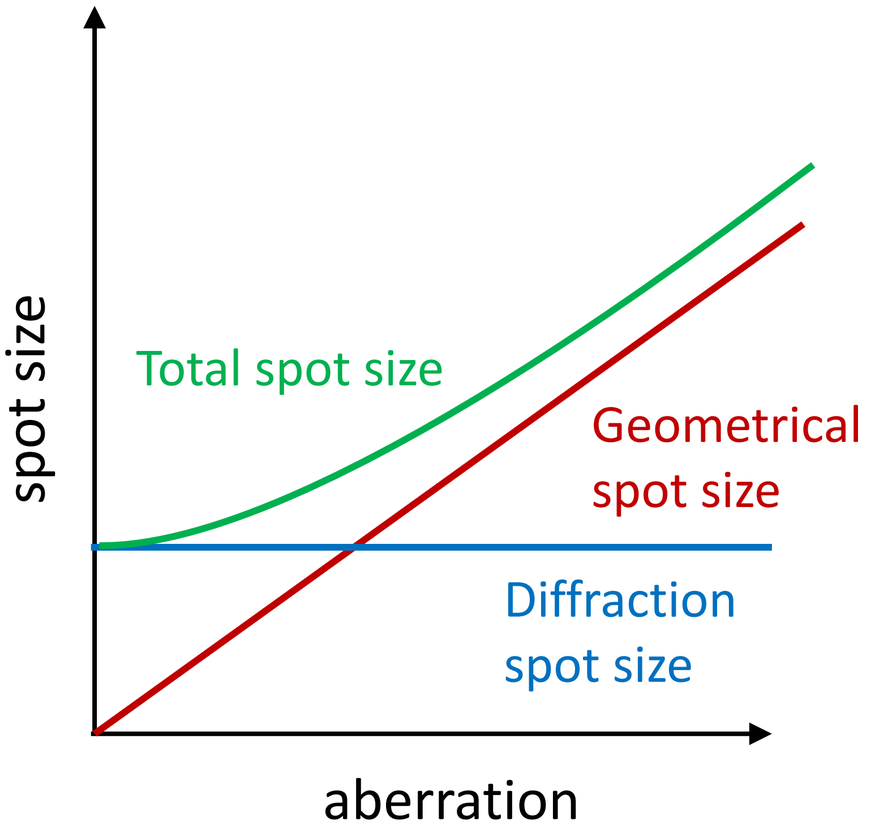




DOWNSCALING OPTICAL SYSTEMS

- **Constant f/# scaling**
  Focal length $f \rightarrow \mathcal{M}f$
  Lens diameter $D \rightarrow \mathcal{M}D$

- Scaling of resolution?

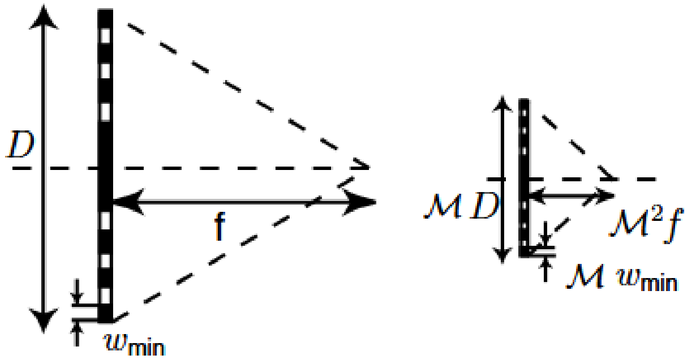


$$\mathrm{SBP_{1D}} = \frac{\Delta x}{\delta x} \approx 1.22 \frac{f}{\lambda}\ \mathrm{FOV} \cdot \mathrm{NA}$$

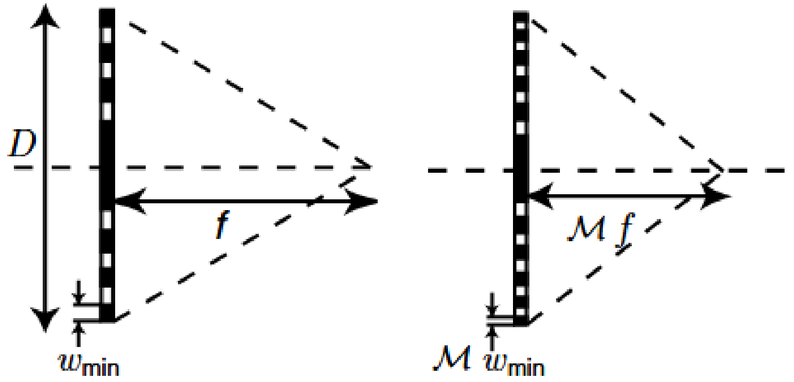
$$\mathrm{SBP} = \frac{\Delta x \Delta y}{\delta x \delta y} \rightarrow \frac{\mathcal{M}^2 \Delta x \Delta y}{\delta x \delta y + \mathcal{M}^2 \sigma^2_{\mathrm{aberration}}} \approx \frac{\mathcal{M}^2 \Delta x \Delta y}{\delta x \delta y}$$

DOWNSCALING OPTICAL SYSTEMS

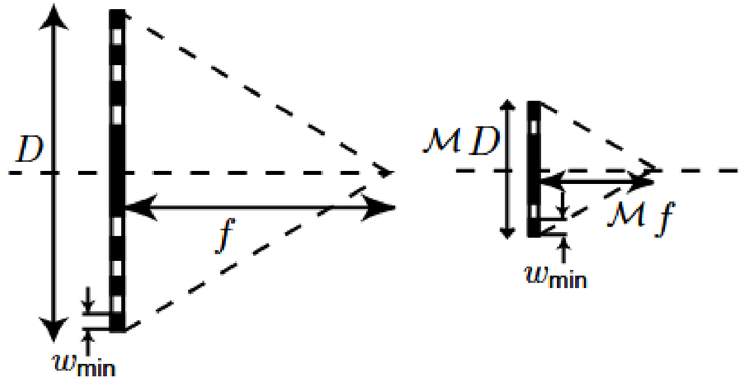
- **Photographic scaling**
  - Focal length $f \rightarrow \mathcal{M}^2 f$
  - Lens diameter $D \rightarrow \mathcal{M} D$
  - Minimum feature size $w_{\min} \rightarrow \mathcal{M} w_{\min}$



- **Microscopic scaling**
  - Focal length $f \rightarrow \mathcal{M} f$
  - Lens diameter $D \rightarrow D$ *(constant)*
  - Minimum feature size $w_{\min} \rightarrow \mathcal{M} w_{\min}$



- **Constant f/# scaling**
  - Focal length $f \rightarrow \mathcal{M} f$
  - Lens diameter $D \rightarrow \mathcal{M} D$
  - Minimum feature size $w_{\min} \rightarrow w_{\min}$ *(constant)*



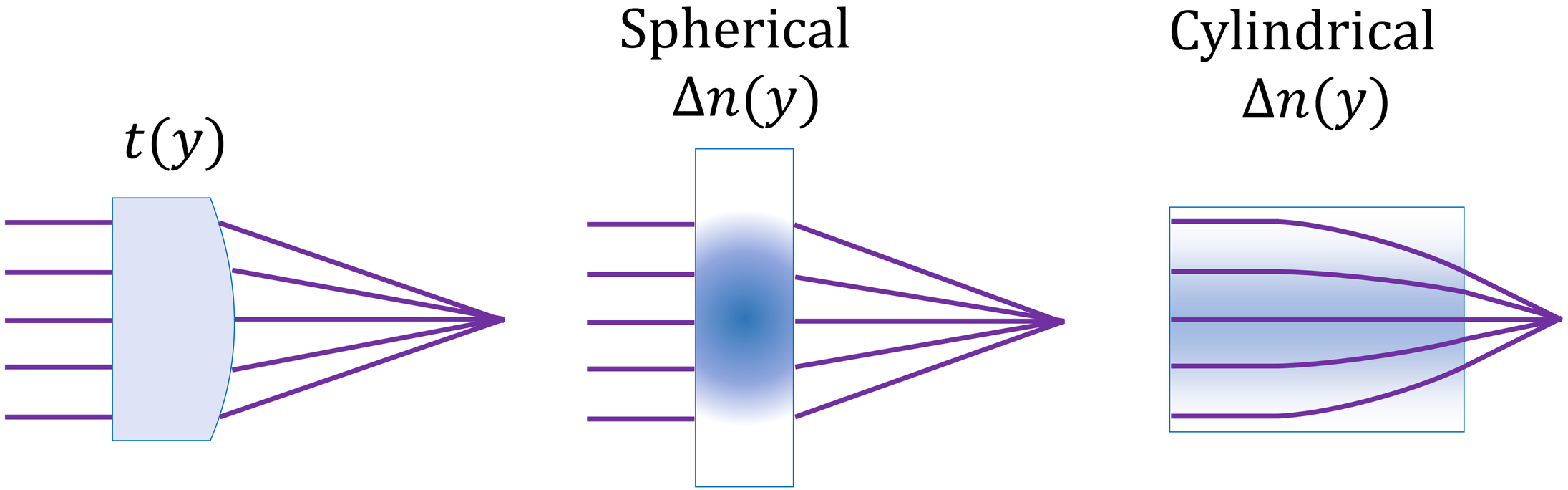
Refractive optics & microlenses

REFRACTIVE ELEMENTS

- Continuous phase difference
- Refractive index Δn and local thickness t

$$\text{Optical path length} \quad \varphi(y) = \frac{2\pi}{\lambda} \Delta n(y)\, t(y)$$

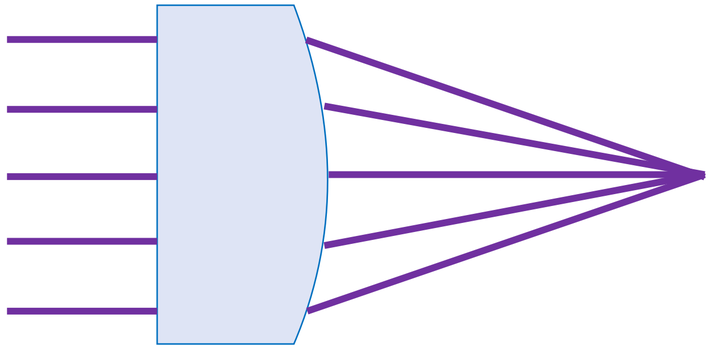
- Same as Snell's law



REFRACTIVE ELEMENTS: MICROLENS

- Fabrication of micro-surfaces:
  - Diamond turning
  - Small diamond tip radii:
  - Shape limited by tip size
    (> microns)

$$t(y)$$

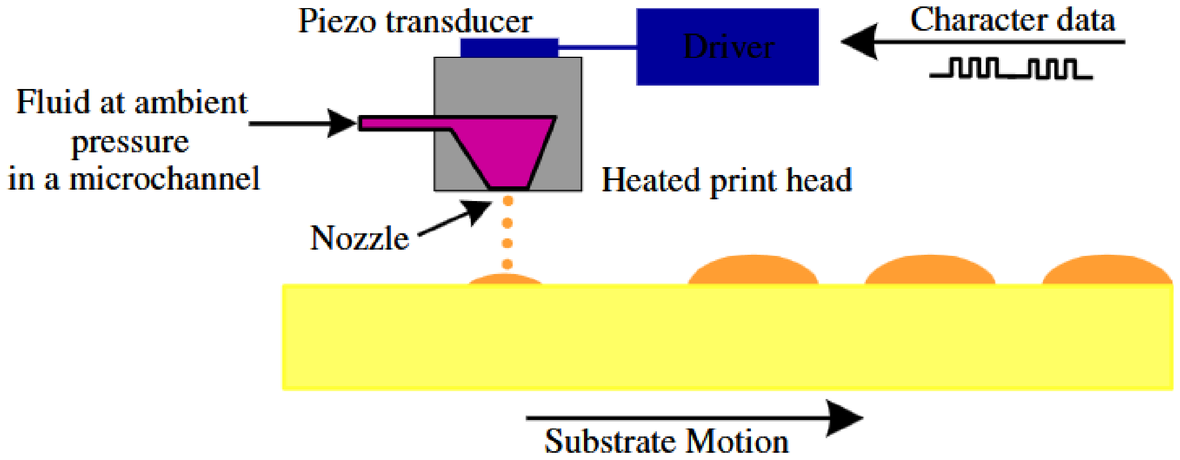


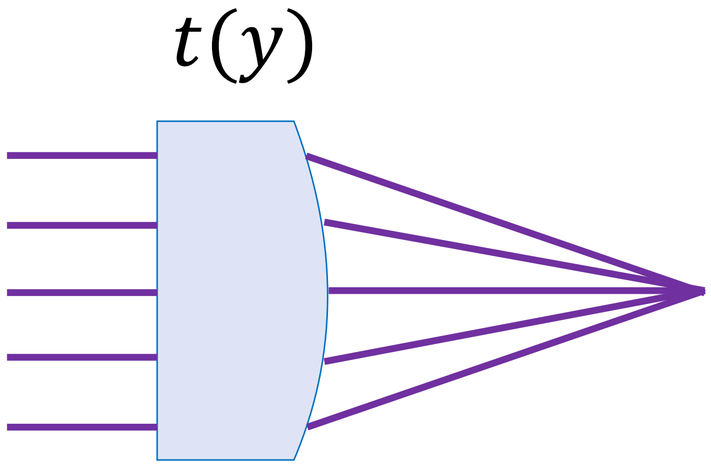


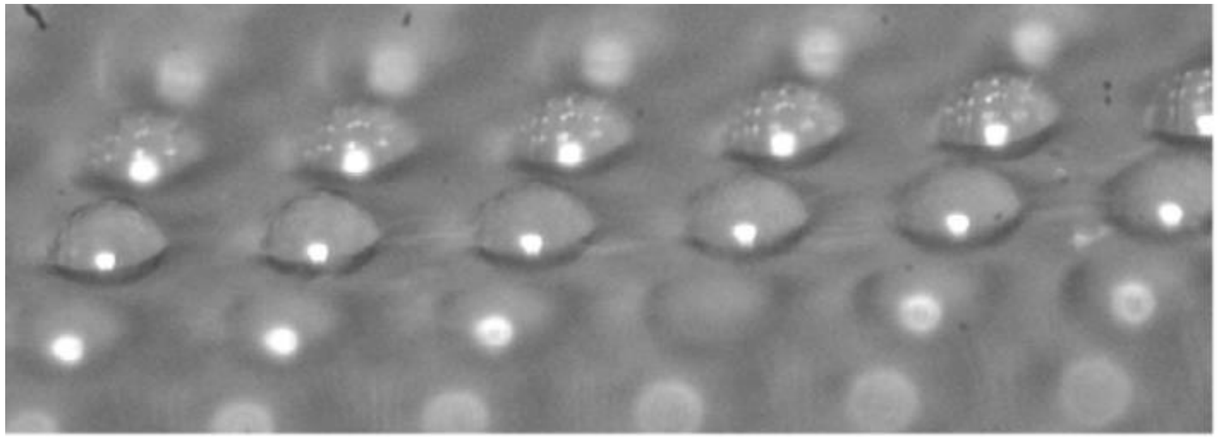
Picture from https://www.precitech.com

REFRACTIVE ELEMENTS: MICROLENS

- Fabrication of micro-surfaces:
  - Inkjet printing of lenses
  - Fast and cost effective

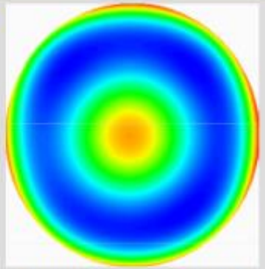
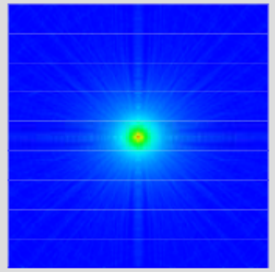
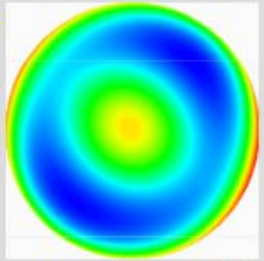
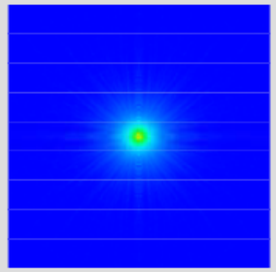
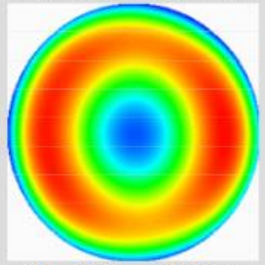
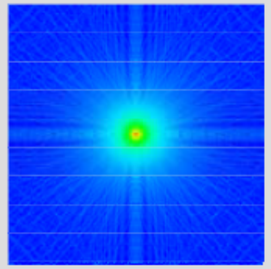
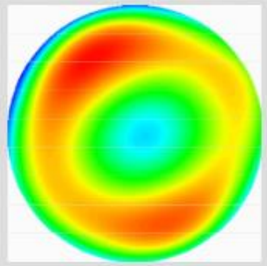
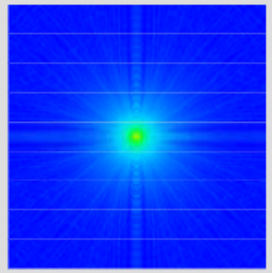




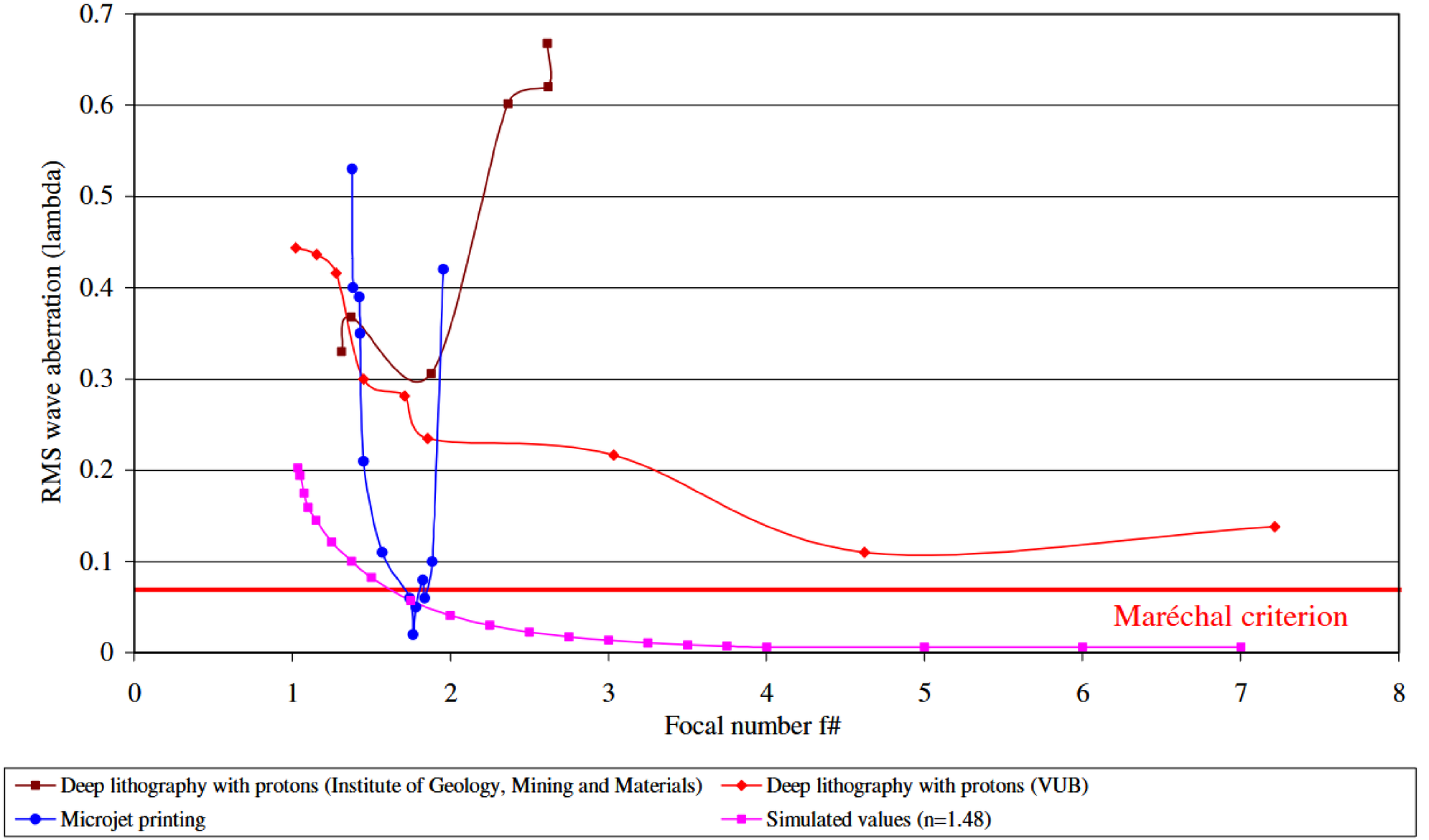


Picture from https://www.precitech.com

EXAMPLE

Microlens fabrication technology	Wavefront aberration	PSF	Strehl ratio	Deviation from an ideal sphere	Surface roughness R_q (50 x 50 µm^2)	Ratio surface roughness / sag
Ion exchange D= 239.9 µm f= 627 µm f#= 2.62 NA= 0.19	 RMS 0.13 λ PV 0.56 λ		0.49			
Ion exchange with subsequent swelling D= 84.4 µm h= 6.8 µm f= 114.9 µm f#= 1.36 NA= 0.36	 RMS 0.08 λ PV 0.38 λ		0.78		6.67 nm	0.09 %
Thermal reflow D= 144.7 µm h= 20.2 µm f= 209 µm f#= 1.44 NA= 0.3445	 RMS 0.09 λ PV 0.38 λ		0.71	RMS 0.0936 λ PV 0.5753 λ	1.87 nm	0.01 %
Reactive ion etching (Heriot-Watt) D= 199.3 µm h= 5.7 µm f= 1809 µm f#= 9.19 NA= 0.0544	 RMS 0.02 λ PV 0.09 λ		0.99	RMS 0.1051 λ PV 0.4425 λ	1.55 nm	0.03 %

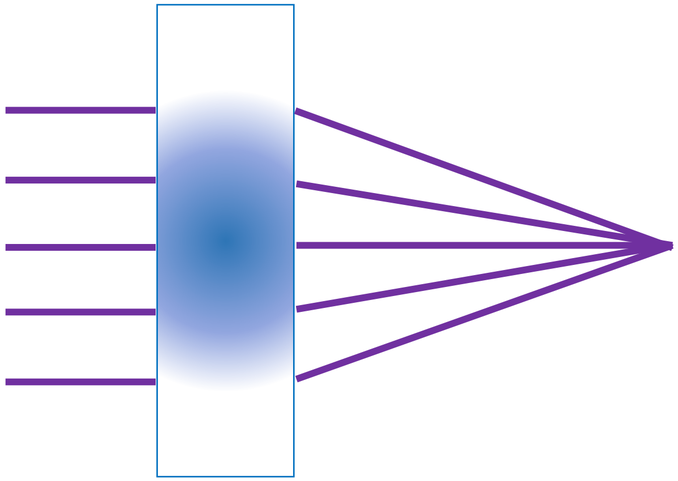
EXAMPLE

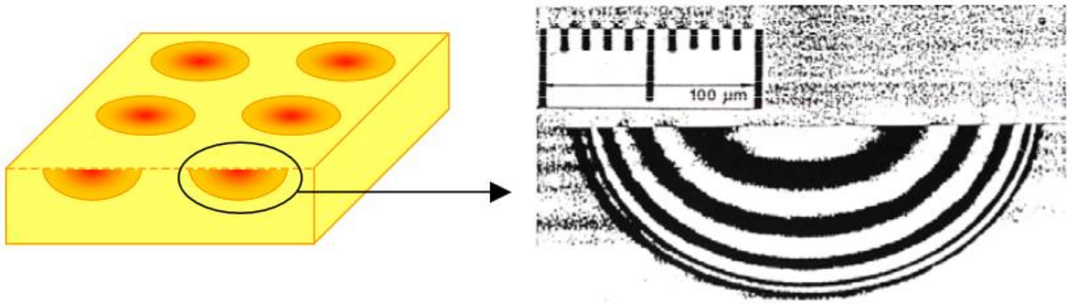

0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
RMS wave aberration (lambda)
0
1
2
3
4
5
6
7
8
Focal number f#
Maréchal criterion
Deep lithography with protons (Institute of Geology, Mining and Materials)
Deep lithography with protons (VUB)
Microjet printing
Simulated values (n=1.48)

REFRACTIVE ELEMENTS: GRIN PLANAR LENS

- Diffusion of positive ions (Ag^+) on planar substrate at “single point”
- Concentration of ions in time: $C(y,t)$
- Diffusion of ions often nonlinear (similar to “aspheric” lens)
- Refractive index difference:

$$\Delta n(y) \propto C(y,t)$$

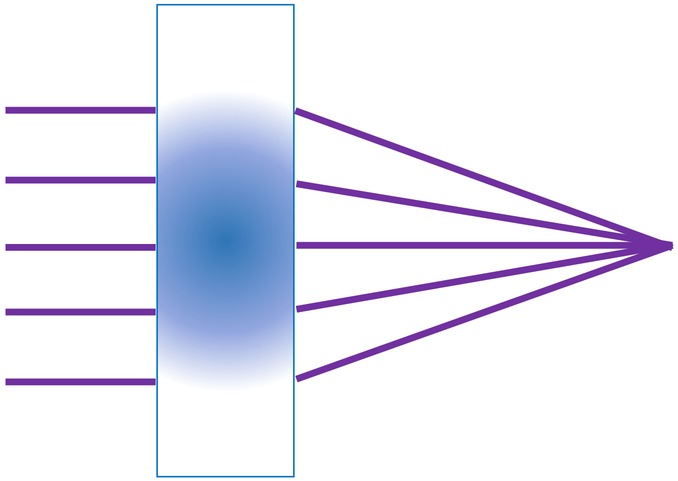




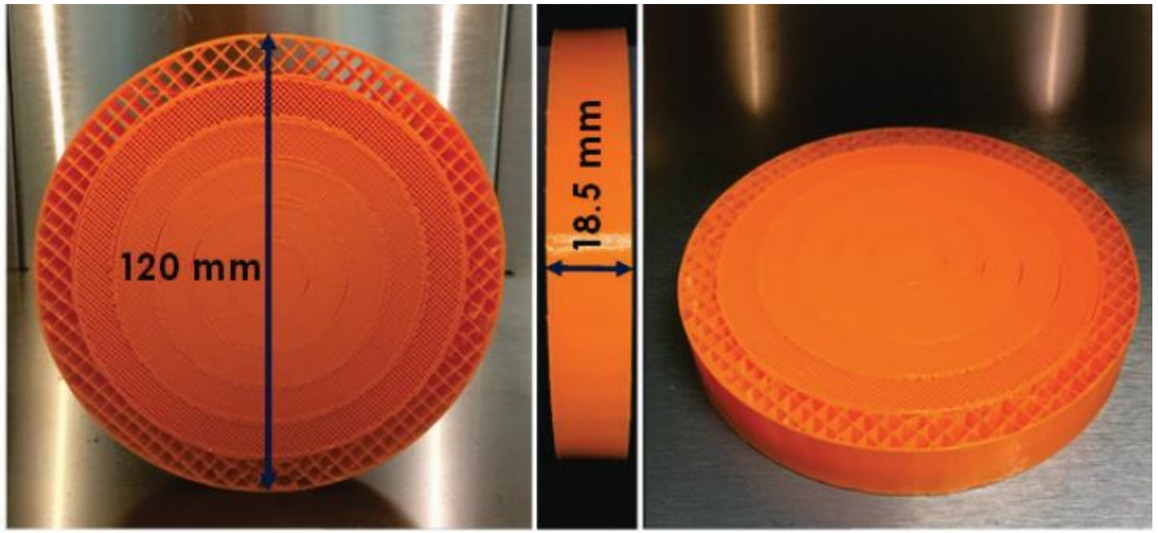
REFRACTIVE ELEMENTS: GRIN PLANAR LENS

- Diffusion of positive ions (Ag^+) on planar substrate at “single point”
- Concentration of ions in time: $C(y,t)$
- Diffusion of ions often nonlinear (similar to “aspheric” lens)
- Refractive index difference:

$$\Delta n(y) \propto C(y,t)$$



Make your own GRIN lens: 3D-printed planar lens (for 13-18 GHz) made from PLA, price: ±10 dollar



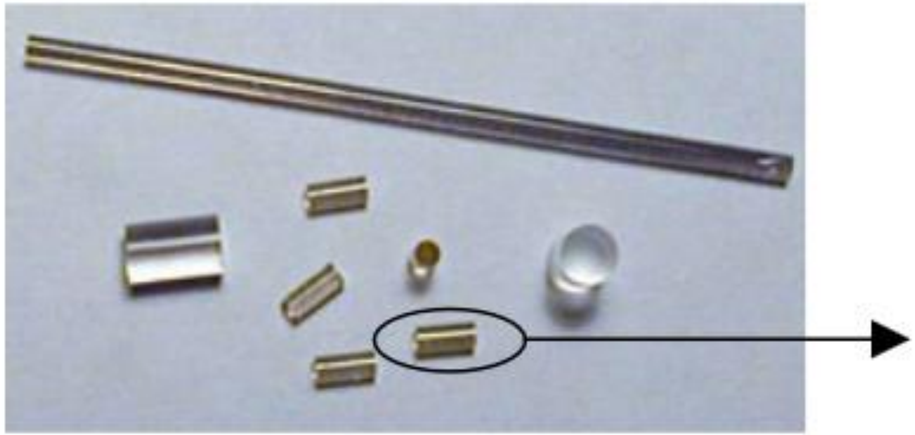
Shiyu Zhang et al. 3D-Printed Graded Index Lens for RF Applications, Proceedings of ISAP2016, Okinawa, Japan

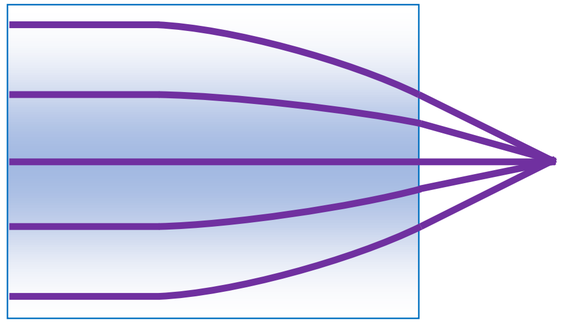
REFRACTIVE ELEMENTS: GRIN ROD LENS

- Diffusion of positive ions (Ag^+) into cylindrical rod
- Concentration of ions in time: $C(r,t)$

$$\Delta n(r) = n_0\left(1 - \frac{1}{2}g^2r^2\right)$$

- Stronger refraction index gradient – strong magnefication

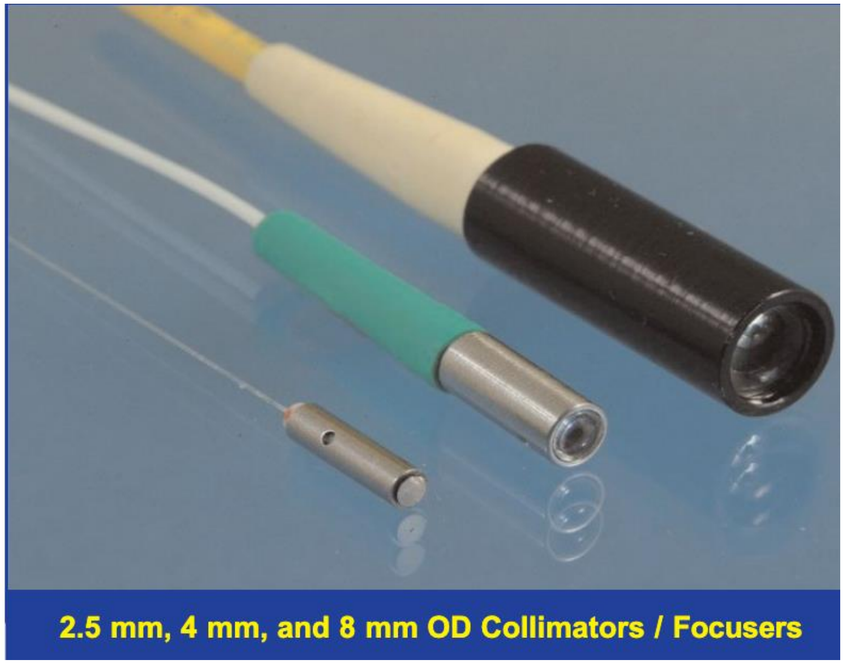




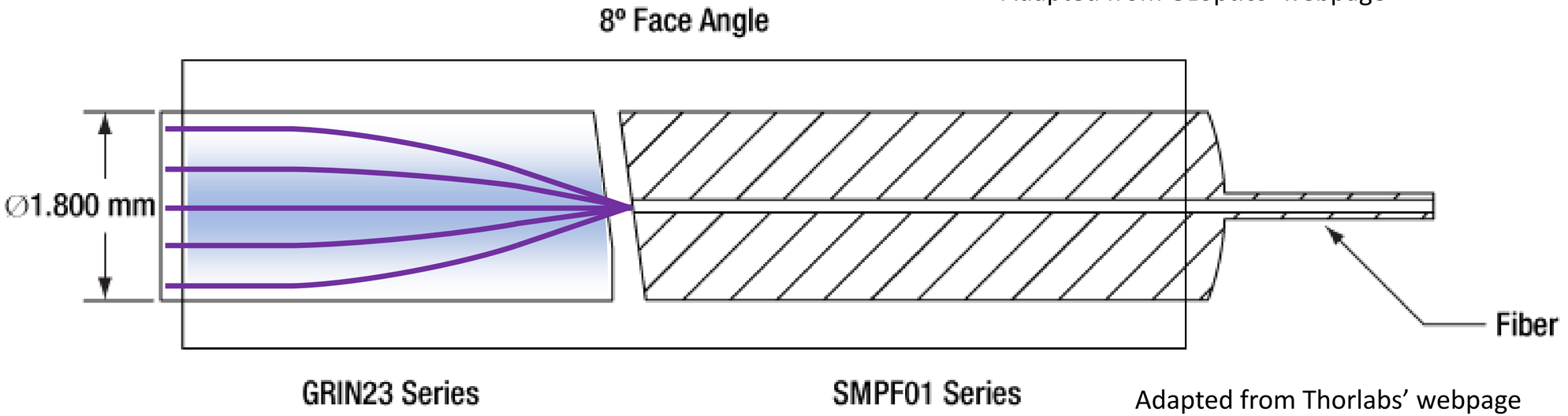
GRIN ROD LENS APPLICATIONS

- Collimation of light from a fiber
- Coupling laser source into fiber
- Coupling (focusing) into waveguide

- Non-normal angle to prevent reflection



Adapted from Ozoptics' webpage



Adapted from Thorlabs' webpage